POC plant · 9 trips · March 2026

Which sensor tripped the plant?

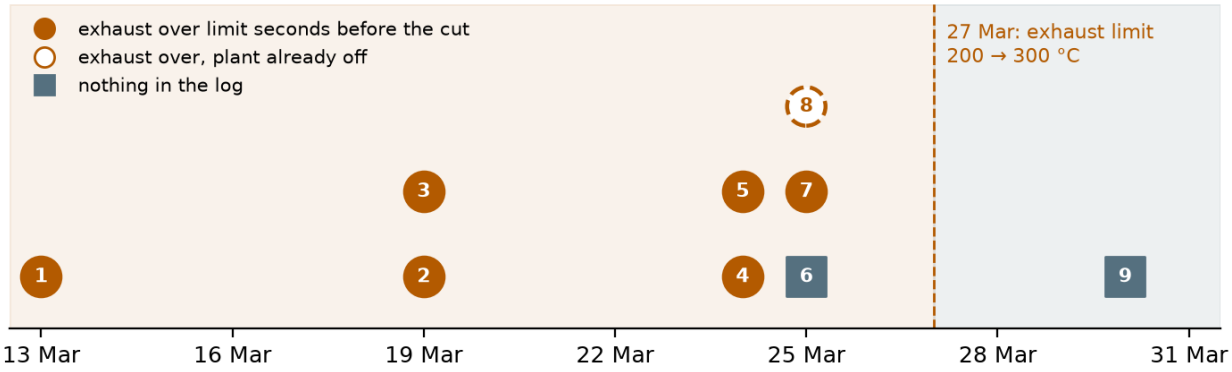
Explainer for the POC SOFC data, 4 Nov 2026. Online version: https://claude.ai/artifact/5AbN9B7AayyNQwnxjktanr

Yes, the data shows a cause for most trips. In **6 of the 9 trips**, one sensor went over its safety limit a few seconds before the plant cut power: the **exhaust outlet temperature** (TIRZ 8.1.10). It went just over its 200 °C limit, by 0.1 to 2 °C.

Why it matters: the trips were not caused by the fuel cell stack or the heaters getting too hot. They came from a limit at the chimney end that was set tight. The plant team raised that limit to 300 °C on 27 March, and no exhaust trip happened after that.

Count	Meaning
6	trips right after the exhaust passed 200 °C
1	exhaust over limit, but the plant was already off load
2	no sensor near a limit: cause not in the log
0	trips from the stack, heater or pressure limits

The 9 trips on one line



Each mark is one trip (an automatic emergency shutdown), numbered in time order.

Trip by trip

#	When	Plant doing	Sensor over limit	Before cut	Plant did	Cause
1	13 Mar 13:45	Part load, 7.2 A	Exhaust 202.0 °C	19 s	Power cut	Exhaust over limit
2	19 Mar 10:49	Just after load, gas on	Exhaust 201.5 °C	22 s	Power cut, emergency stop 15 min later	Exhaust over limit
3	19 Mar 15:54	Part load, 6.9 A	Exhaust 200.9 °C	18 s	Power cut	Exhaust over limit
4	24 Mar 09:25	Part load, 13.8 A	Exhaust 200.1 °C	25 s	Emergency purge, emergency stop 17 min later	Exhaust over limit
5	24 Mar 14:36	Part load, 7.2 A	Exhaust 200.5 °C	4 s	Emergency purge, emergency stop 2 min later	Exhaust over limit
6	25 Mar 08:50	Part load, 11.8 A	None (exhaust 108 °C)	–	Emergency stop	Nothing in the log
7	25 Mar 10:58	Hot standby, just after load	Exhaust 200.1 °C	5 s	Power cut (heater)	Exhaust over limit

#	When	Plant doing	Sensor over limit	Before cut	Plant did	Cause
8	25 Mar 15:28	Load already off	Exhaust 202.7 °C, over for 6 min	not the trigger	Emergency stop	Exhaust over, plant already off
9	30 Mar 09:30	Fuel switch start, 0 A	None (exhaust 119 °C)	–	Emergency stop	Nothing in the log

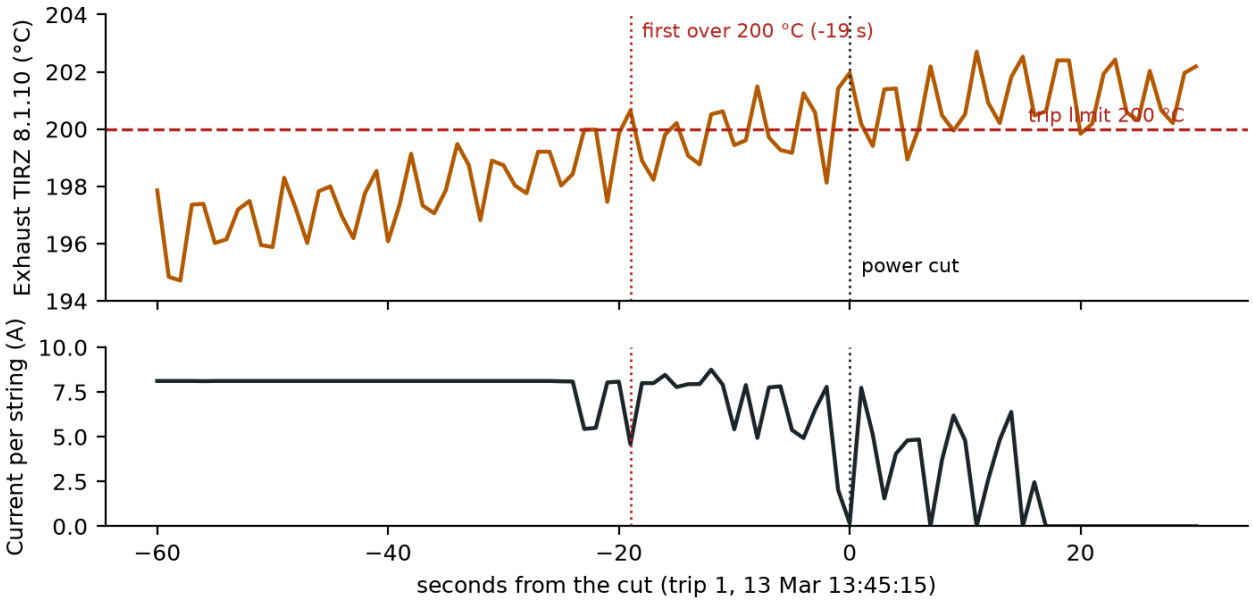
"Power cut" = stack current and/or air heater switched off, gas valves unchanged. "Emergency purge" = the safety controller also switched the gas to the nitrogen + hydrogen flush. "Emergency stop" = all gas valves closed.

If you remember one thing: 6 of 9 trips were set off by the exhaust temperature going a hair over 200 °C. The stack itself never reached its 720 °C safety limit.

A limit set at the chimney end

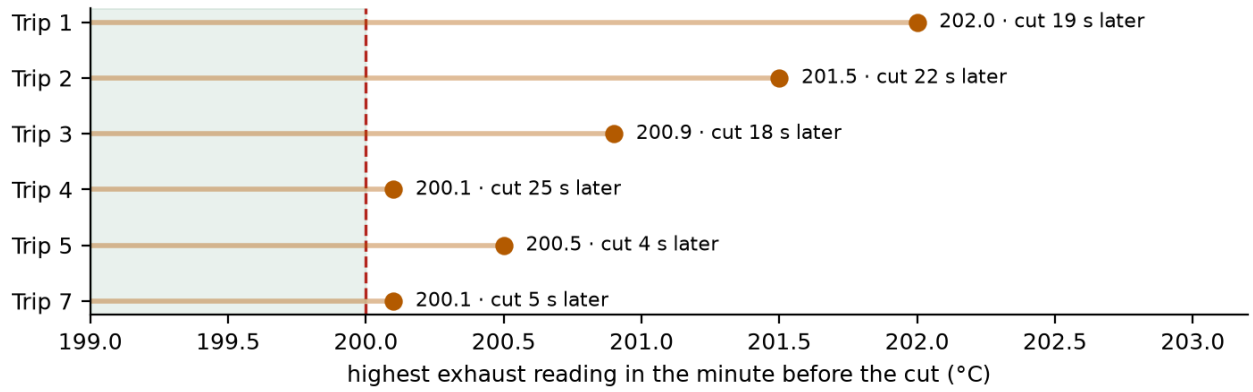
The exhaust sensor TIRZ 8.1.10 measures the gas leaving the plant, after it has given its heat back in the heat exchangers. In part load this gas ran at about 196–200 °C, right up against the 200 °C safety limit.

One trip second by second



Trip 1, 13 March, one reading per second. The exhaust reading wobbles by about ±2 °C. It first passes 200 °C 19 seconds before the cut. Four seconds later the current starts to fall, and at the cut it is zero.

Each exhaust trip went over by a hair



Every reading is 0.1 to 2 °C over the limit, about the size of the sensor's normal wobble.

Check	Result
Times the exhaust passed the limit while the plant was producing power or heating	7 (in 6 trips)
Of those, followed by a power cut or purge within 25 s	7 of 7
Times the exhaust passed 200 °C with no power or heating on (idle, or already cut)	13, no cut (nothing left to cut)
Exhaust above 200 °C while running on 30 Mar, after the limit went to 300 °C	116 s, no trip

How the plant reacted

• **Trips 4 and 5:** the full safety response written in the plant's safety rules: power off, gas switched to the protective nitrogen + hydrogen flush (emergency purge).

• **Trips 1, 2, 3, 7:** only the power was cut, and the gas valves stayed open. This is softer than the written safety response, so the cut probably came from the normal control system or the operator reacting to the alarm. The log cannot tell which.

• In trips 2, 4 and 5 an emergency stop followed 2–17 minutes later while the exhaust was still over 200 °C.

• The cut came 4–25 s after the logged reading first passed 200 °C. The safety controller is meant to act after 0.05 s, so either it reads its own, slightly different signal, or the cut was not instant. The log has one value per second and no "trip reason" column.

If you remember one thing: every time the exhaust passed 200 °C while the plant was running, power was cut within 25 seconds. After the limit was raised to 300 °C, it stopped happening.

Three trips with no sensor to blame

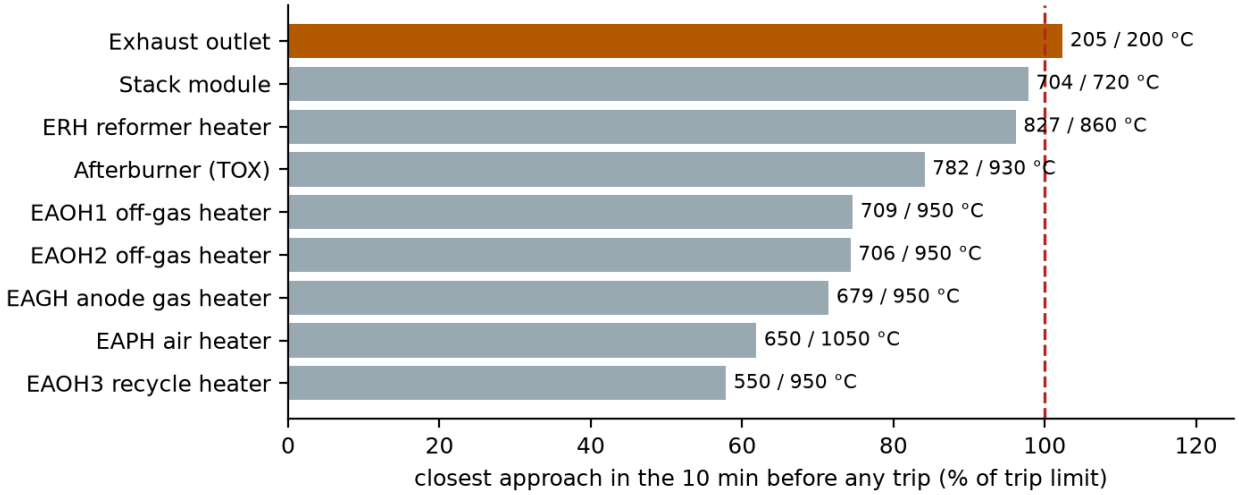
Trips 6, 8 and 9 were emergency stops: every gas valve closed at once. Before trips 6 and 9 all readings were calm and steady. Nothing was near a limit.

What can set off an emergency stop

The plant's safety rules list: gas detectors (O2, H2, CO in the containers), smoke detectors, emergency stop buttons and loss of exhaust fans. **None of these is recorded in the data log**; only the result (all gas valves shut, air and power off) is visible.

Trip	What we see	Most likely
6 · 25 Mar 08:50	Part load at 11.8 A, stack 658 °C, exhaust 108 °C, all steady for 90 s, then every gas valve shut in one step	Operator button or a gas/smoke detector; cannot tell
8 · 25 Mar 15:28	Load already off. Exhaust had been over 200 °C for 6 minutes with no reaction	Operator stop during cooling (exhaust was not the direct trigger)
9 · 30 Mar 09:30	Start of fuel switch, 0 A, stack 608 °C, exhaust 119 °C, all steady	Operator button or a gas/smoke detector; cannot tell

No other safety sensor came close



Highest reading of each temperature safety sensor in the 10 minutes before any trip, as a share of its limit. Next closest after the exhaust: stack 704 °C (16 °C short) and reformer heater 827 °C (33 °C short). The low-pressure guard PIRZ 8.1.2 never fell below 0.94 bar (trip below 0.75).

Two other checks found nothing. The weakest-cell voltage did drop below its setting, but it does that in 38 % of all running seconds without any trip. It is a target the power controller steers to, not a trip. The insulation readings did not change at any trip.

What would close the gap

• Ask the plant team for the **safety controller's alarm and event list** for 13–30 March. It names the trigger of each trip.

• Ask whether trips 6 and 9 were **operator stops**. Both happened during load tests.

• For future logging, add a **trip reason signal** and the gas-detector readings to the data logger.

How this was checked: all 9 trips from the trip list, raw 1-second data, every logged safety sensor against the limits in the plant's safety function list (27 Mar 2026). Code: data_analysis/scripts/trip_check.py plus a one-off check script.

If you remember one thing: two trips (6 and 9) have no cause in the data at all. Only the plant's own safety alarm list can explain them.